

# Employee Attrition Prediction Using Machine Learning

## Dataset

IBM HR Analytics Employee Attrition Dataset

<https://www.kaggle.com/datasets/pavansubhasht/ibm-hr-analytics-attrition-dataset>

## Target

Predict whether an employee will **Leave** or **Stay** in the company.

## Tasks

1. Load and understand the dataset.
2. Check missing values and duplicates.
3. Perform EDA and visualization.
4. Analyze Attrition with Age, Salary, Overtime, Job Satisfaction, Work-Life Balance, Department, Job Role and Years at Company.
5. Encode categorical data and scale features where required.
6. Split data into 80% training and 20% testing.
7. Train:
  - Logistic Regression
  - KNN
  - Decision Tree
  - Random Forest
  - Gradient Boosting
8. Compare models using Accuracy, Precision, Recall and F1 Score.
9. Select the best model.
10. Create a Confusion Matrix.
11. Find Top 5 important features affecting attrition.
12. Predict Attrition for a new employee.
13. Show:
  - Leave / Stay
  - Attrition Probability
  - Low / Medium / High Risk
14. Give 5 HR recommendations based on the analysis.
15. Write a short final conclusion.

## Final Output

The project should take new employee details and display:

**Prediction:** Leave / Stay

**Attrition Probability:** XX%

**Risk Level:** Low / Medium / High